

Quiz 3

1. ROC and DTFT

$X(z) = 1/(1 - 1.25z^{-1})$ with ROC $|z| < 1.25$. Does the DTFT exist? Is the corresponding sequence right-sided or left-sided?

Pole at $z = 1.25$, ROC $|z| < 1.25$ implies left-sided sequence and unit circle in ROC => DTFT exists.

2. Inverse z-transform

Find $x[n]$ for $X(z) = 1/(1 - 0.6z^{-1})$, ROC $|z| > 0.6$.

$x[n] = (0.6)^n u[n]$

3. Transfer function

For $y[n] - 0.5y[n-1] = x[n] - x[n-1]$, find $H(z)$, its pole and its zero.

$H(z) = \frac{1-z^{-1}}{1-0.5z^{-1}} = \frac{z-1}{z-0.5}$ zero: $z = 1$, pole: $z = 0.5$.

4. Causal and stable rule

A causal rational system has a pole at $z=1.02$. Can it be BIBO stable? Briefly justify.

For a causal system ROC is outside the outermost pole, here ROC $|z| > 1.02$. This does not include unit circle => system cannot be BIBO stable.

5. Geometry

What happens to $|H(e^{j\omega})|$ at a frequency whose unit-circle point coincides exactly with a zero?

What happens qualitatively near a pole that lies very close to the unit circle?

At a zero on the unit circle a numerator distance of $|H(e^{j\omega})|$ is zero so $|H(e^{j\omega})| = 0$ (perfect notch). Near a pole close to the unit circle, a denominator distance of $|H(e^{j\omega})|$ is small, so $|H(e^{j\omega})|$ becomes large (resonant peak).